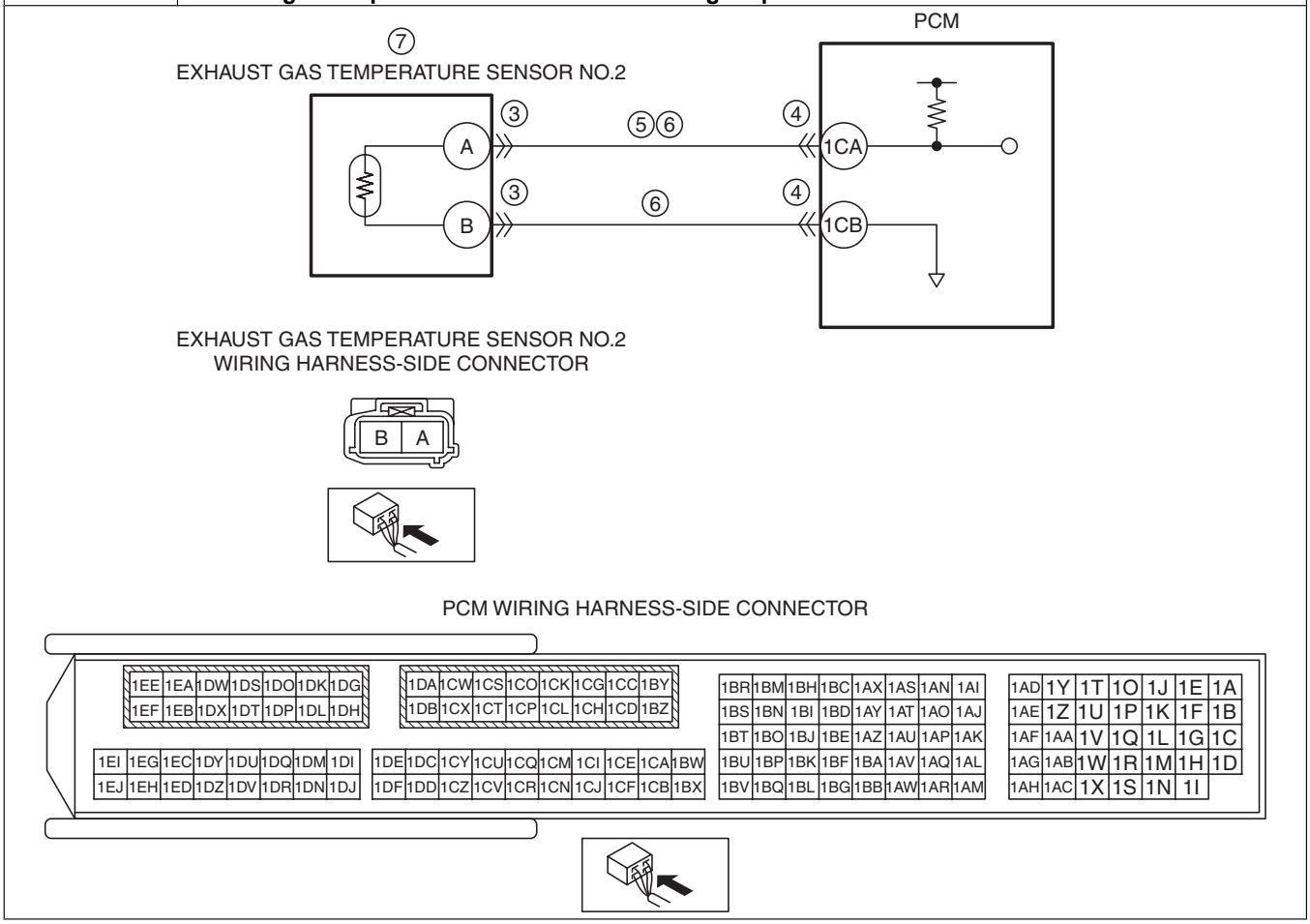
DTC P2033:00 [SKYACTIV-D 1.5]

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DTC P2033:00	Exhaust gas temperature sensor No.2 circuit high input
DETECTION CONDITION	• The PCM monitors the exhaust gas temperature sensor No.2 signal. If the PCM detects that the exhaust gas temperature sensor No.2 voltage at the PCM terminal 1CA is **above 4.96 V** for **10 s**, the PCM determines that the exhaust gas temperature sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — Between the elapsed time of **14 to 70 min.** after the ignition is switched on (engine on). — Engine speed: **above 700 rpm** — Engine coolant temperature: **above 60 °C {140 °F}** — Charge air temperature: above **-10 °C {14 °F}** — Intake air temperature: above **-10 °C {14 °F}** — Vehicle speed: **above 25 km/h {16 mph}** continues for **10 min or more** — Fuel injection amount: **16 mm^3/st or more** — Total fuel injection amount: **250,000 mm^3 or more** — IAT sensor No.2: Normal — The following DTCs are not detected: • IAT sensor No.1: P0112:00, P0113:00 • ECT sensor: P0117:00, P0118:00 • IAT sensor No.2: P007C:00, P007D:00 • VSS: P0500:00 **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA /Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Exhaust gas temperature sensor No.2 connector or terminals malfunction • PCM connector or terminals malfunction • Short to power supply in wiring harness between exhaust gas temperature sensor No.2 terminal A and PCM terminal 1CA • Open circuit in wiring harness between the following terminals: — Exhaust gas temperature sensor No.2 terminal A—PCM terminal 1CA — Exhaust gas temperature sensor No.2 terminal B—PCM terminal 1CB • Exhaust gas temperature sensor No.2 malfunction • PCM malfunction

DTC P2033:00	Exhaust gas temperature sensor No.2 circuit high input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT EXHAUST GAS TEMPERATURE SENSOR NO.2 CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the exhaust gas temperature sensor No.2 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT EXHAUST GAS TEMPERATURE SENSOR NO.2 SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the exhaust gas temperature sensor No. 2 and PCM connectors are disconnected. • Switch the ignition ON (engine off). • Measure the voltage at the exhaust gas temperature sensor No.2 terminal A (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between exhaust gas temperature sensor No.2 terminal A and PCM terminal 1CA. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 8.
6	**INSPECT EXHAUST GAS TEMPERATURE SENSOR NO.2 CIRCUIT FOR OPEN CIRCUIT** • Verify that the exhaust gas temperature sensor No. 2 and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side): — Exhaust gas temperature sensor No.2 terminal A—PCM terminal 1CA — Exhaust gas temperature sensor No.2 terminal B—PCM terminal 1CB • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust gas temperature sensor No.2 terminal A—PCM terminal 1CA • Exhaust gas temperature sensor No.2 terminal B—PCM terminal 1CB **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.
7	**INSPECT EXHAUST GAS TEMPERATURE SENSOR NO.2** • Reconnect all disconnected connectors. • Inspect the exhaust gas temperature sensor No.2. (See EXHAUST GAS TEMPERATURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas temperature sensor No.2, then go to the next step. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Access EXHTEMP2 PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Start the engine and idle it **1 minutes**. • Confirm that the EXHTEMP2 PID is **–20 °C {–4 ° F}** or more. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.